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BASIC MATHEMATICS

FORM FOUR

Certificate of Secondary Education Examination (CSEE) 2024 Answer Key and Worked Solutions

Subject	Basic Mathematics
Code	041
Level	Form Four
Year	2024
Exam Board	NECTA
Exam	Certificate of Secondary Education Examination (CSEE)
Type	Answer key and worked solutions
Questions	14 compulsory questions; 40 answered parts

Try each question before checking the solution. Follow the steps, compare your working, and practise the parts you find difficult.

Source paper: BasicMath-F4-2024.pdf

Question 9(b) does not specify the order of the ground attachment points. Both possible collinear wire lengths are explained. The inverse in question 14(a) is interpreted on nonnegative question counts.

Section A — Worked solutions

1(a). Find the largest equal squares that tile a board 54 cm by 78 cm.

Step 1: The largest side must divide both board dimensions, so find $\text{GCF}(54, 78)$.

Step 2: $54 = 2 \times 3^3$ and $78 = 2 \times 3 \times 13$; common factor $= 2 \times 3 = 6$. The board fits 9×13 such squares.

Final Answer: Each square has side 6 cm (area 36 cm²).

1(b)(i). Of 40 students, 17 are boys and the rest girls. Find both percentages.

Step 1: Girls $= 40 - 17 = 23$.

Step 2: Girls percentage $= (23/40) \times 100 = 57.5\%$; boys $= (17/40) \times 100 = 42.5\%$.

Final Answer: Girls: 57.5%; boys: 42.5%.

1(b)(ii). Express the boys' share of the class as a decimal.

Step 1: The fraction of boys is $17/40$.

Step 2: Multiply top and bottom by 25: $17/40 = 425/1000 = 0.425$.

Final Answer: 0.425 of the class (17 boys).

2(a). Express $(5 + \sqrt{7})/(3 + \sqrt{7})$ as $a + b\sqrt{c}$ with integer a,b,c.

Step 1: Multiply by conjugate $(3 - \sqrt{7})/(3 - \sqrt{7})$.

Step 2: Numerator $= 15 - 5\sqrt{7} + 3\sqrt{7} - 7 = 8 - 2\sqrt{7}$; denominator $= 9 - 7 = 2$. Divide by 2.

Final Answer: $4 - \sqrt{7}$; $a = 4$, $b = -1$, $c = 7$.

2(b). Without tables or a calculator, solve $3 \log_3 x + 4 = \log_3 24$.

Step 1: For $x > 0$, use $4 = \log_3 81$ and $3 \log_3 x = \log_3 x^3$.

Step 2: $\log_3(81x^3) = \log_3 24$ gives $x^3 = 24/81 = 8/27$; take the positive cube root.

Final Answer: $x = 2/3$.

3(a). Of 100 students, 45 prefer Music, 40 Theatre Arts and 5 both. Use a formula to find those preferring neither.

Step 1: $n(M \cup T) = n(M) + n(T) - n(M \cap T) = 45 + 40 - 5 = 80$.

Step 2: Neither = $100 - 80 = 20$.

Final Answer: 20 students.

3(b)(i). Among 20 offices, table counts 1,2,3,4,5,6 occur in 4,5,6,2,1,2 offices respectively. Find P(two tables).

Step 1: Five offices have exactly two tables.

Step 2: Probability = favourable offices/total offices = $5/20 = 1/4$.

Final Answer: 1/4.

3(b)(ii). Find P(at least five tables).

Step 1: At least five includes five or six tables.

Step 2: Favourable offices = $1 + 2 = 3$, so probability = $3/20$.

Final Answer: 3/20.

4(a). A(6,−4),B(8,4),P(6,1),Q(7,5): determine whether AB ∥ PQ.

Step 1: $m_{AB} = [4 - (-4)]/(8 - 6) = 8/2 = 4$.

Step 2: $m_{PQ} = (5 - 1)/(7 - 6) = 4$. Equal gradients give parallel lines; the lines are distinct because their intercepts differ.

Final Answer: Yes, AB ∥ PQ.

4(b). $a = 2i - 7j$ and $2a + 3b = 13i - 2j$. Find b.

Step 1: $2a = 4i - 14j$. Subtract from the stated equation: $3b = (13 - 4)i + (-2 + 14)j = 9i + 12j$.

Step 2: Divide both components by 3.

Final Answer: $b = 3i + 4j$, or (3,4).

5(a)(i). A regular hexagonal plot of area 720 m^2 is inscribed in a circle of radius r . Find r to the nearest tens.

Step 1: Six equilateral triangles of side r have total area $6(\sqrt{3}/4)r^2 = (3\sqrt{3}/2)r^2$.

Step 2: Set $(3\sqrt{3}/2)r^2 = 720$, giving $r^2 = 480/\sqrt{3} = 160\sqrt{3}$ and $r \approx 16.6472 \text{ m}$.

Step 3: Round to the nearest ten metres: 16.6472 is closer to 20 than to 10.

Final Answer: 20 m.

5(a)(ii). Give the same radius to the nearest tenth.

Step 1: $r = \sqrt{160\sqrt{3}} \approx 16.6472 \text{ m}$.

Step 2: Keep one decimal place; the next digit is 4, so the tenth remains 6.

Final Answer: 16.6 m.

5(b)(i). In the plot diagram, A and D are right angles, A,E,D are collinear and B,E,C are collinear. Prove the triangles similar.

Step 1: $\angle BAE = \angle CDE = 90^\circ$.

Step 2: $\angle AEB = \angle DEC$ because they are vertically opposite. Two equal corresponding angles give similarity; $A \leftrightarrow D$, $E \leftrightarrow E$, $B \leftrightarrow C$.

Final Answer: $\triangle AEB \sim \triangle DEC$ by AA.

5(b)(ii). $AB = 36 \text{ m}$, $AE = 24 \text{ m}$ and $ED = 32 \text{ m}$. Find CD .

Step 1: By similarity, $CD/AB = DE/AE = 32/24 = 4/3$.

Step 2: $CD = 36 \times 4/3 = 48 \text{ m}$.

Final Answer: 48 m.

6(a). 850 ml rainwater is collected per minute from 8:10 a.m. to 11:52 a.m. Find the litres collected.

Step 1: Elapsed time = 3 hours 42 minutes = 222 minutes.

Step 2: Volume = $850 \times 222 = 188,700 \text{ ml} = 188.7 \text{ litres}$.

Final Answer: 188.7 litres.

6(b)(i). Mary's travel time t seconds varies inversely with speed v m/s; 30 minutes at 10 m/s. Write t in terms of v .

Step 1: $t = k/v$. Convert 30 minutes to 1800 seconds.

Step 2: $k = tv = 1800 \times 10 = 18,000$, giving the required equation.

Final Answer: $t = 18,000/v$.

6(b)(ii). What average speed gets Mary home in 15 minutes?

Step 1: 15 minutes = 900 seconds.

Step 2: $900 = 18,000/v$ gives $v = 18,000/900 = 20$ m/s.

Final Answer: 20 m/s.

7(a). The sum and difference of Amina's and Bakar's ages are in ratio 5:4. Find the ratio of their ages.

Step 1: Let $A + B = 5k$ and $A - B = 4k$. Add: $2A = 9k$; subtract: $2B = k$.

Step 2: $A/B = (9k/2)/(k/2) = 9$.

Final Answer: 9:1, taking Amina as the older person as in $A - B$.

Study Tip: If Bakar is the older person instead, the respective Amina:Bakar ratio is 1:9. The larger-age:smaller-age ratio is always 9:1.

7(b). Extract the trial balance from the July cash account.

Step 1: Purchases = $700,000 + 300,000 + 235,000 = 1,235,000$. Sales = $500,000 + 310,000 = 810,000$.

Step 2: Purchases, expenses and cash carry debit balances; sales and capital carry credit balances.

Step 3: Check debit sum $1,235,000 + 110,000 + 55,000 + 410,000 = 1,810,000$; credit sum $1,000,000 + 810,000 = 1,810,000$.

Final Answer: Debits: Purchases 1,235,000; Rent 110,000; Wages 55,000; Cash 410,000. Credits: Capital 1,000,000; Sales 810,000. Each total = Tsh 1,810,000.

8(a). Monthly loan instalments are Tsh 20,000,22,000,24,000,..., ending with 114,000. How long was repayment?

Step 1: This AP has first term 20,000 and difference 2,000.

Step 2: Final term: $114,000 = 20,000 + (n - 1)2,000$. Hence $n - 1 = 47$ and $n = 48$.

Final Answer: 48 months (4 years).

8(b). Find the total loan repaid by those instalments.

Step 1: Sum of n AP instalments $= (n/2)(\text{first} + \text{last})$.

Step 2: $S = (48/2)(20,000 + 114,000) = 24 \times 134,000 = 3,216,000$.

Final Answer: Tsh 3,216,000.

9(a). Show that $4 \sin 75^\circ = \sqrt{6} + \sqrt{2}$.

Step 1: Use $\sin(45^\circ + 30^\circ) = \sin 45^\circ \cos 30^\circ + \cos 45^\circ \sin 30^\circ$.

Step 2: $\sin 75^\circ = (\sqrt{2}/2)(\sqrt{3}/2) + (\sqrt{2}/2)(1/2) = (\sqrt{6} + \sqrt{2})/4$. Multiply both sides by 4.

Final Answer: $4 \sin 75^\circ = \sqrt{6} + \sqrt{2}$.

9(b). A vertical post AB is 6 m high; wire AC is 10 m and ground points C,D are 4 m apart. Find wire AD in surd form.

Step 1: In right triangle ABC, $BC = \sqrt{10^2 - 6^2} = 8$ m.

Step 2: If ground order is B,C,D, then $BD = 8 + 4 = 12$ and $AD = \sqrt{6^2 + 12^2} = \sqrt{180} = 6\sqrt{5}$.

Step 3: If order is B,D,C, $BD = 8 - 4 = 4$ and $AD = \sqrt{6^2 + 4^2} = \sqrt{52} = 2\sqrt{13}$.

Final Answer: If D is 4 m farther from the base than C: $AD = 6\sqrt{5}$ m. If D is 4 m closer: $AD = 2\sqrt{13}$ m. The point order is needed to choose one.

Study Tip: The question gives no diagram or ground-point order. These are the two collinear placements consistent with the stated lengths; neither may be selected without an additional condition.

10(a). A candidate needs an average of at least 53 in two tests and scores 54 in the first. Find the least second-test mark.

Step 1: Let the second mark be x : $(54 + x)/2 \geq 53$.

Step 2: $54 + x \geq 106$ gives $x \geq 52$; the least value is 52.

Final Answer: 52 marks.

10(b). A rectangular garden has area 171 m^2 and its fence uses 56 m of wire. Find its dimensions.

Step 1: Perimeter $2(l + w) = 56$ gives $l + w = 28$; also $lw = 171$.

Step 2: Set $w = 28 - l$: $l(28 - l) = 171$, so $l^2 - 28l + 171 = 0$.

Step 3: Factor $(l - 19)(l - 9) = 0$. The two dimensions are 19 and 9; check $19 \times 9 = 171$ and $2(19 + 9) = 56$.

Final Answer: 19 m by 9 m.

Section Summary: Use dimensions, ratios and rates with consistent units. A complement subtracts a union from the universal total; inverse variation keeps time $\times$ speed constant.

Section B — Worked solutions

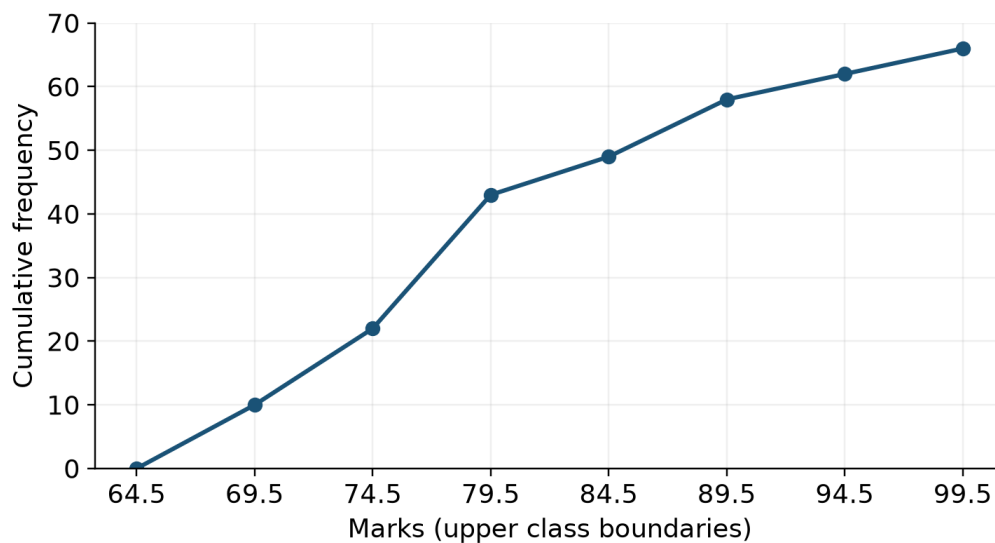
11(a)(i). Classes 65-69,...,95-99 have cumulative frequencies 10,22,43,49,58,62,66. Find the mean using assumed mean 77, to 2 decimal places.

Step 1: Subtract consecutive cumulative totals to obtain frequencies 10,12,21,6,9,4,4. Midpoints are 67,72,77,82,87,92,97.

Step 2: With $u = (\text{midpoint} - 77)/5$, values are $-2, -1, 0, 1, 2, 3, 4$. Products fu are $-20, -12, 0, 6, 18, 12, 16$; sum = 20.

Step 3: Mean = $77 + 5(20/66) = 78.51515\dots$, rounding to 78.52.

Final Answer: 78.52 marks.

11(a)(ii). Draw the cumulative-frequency curve (ogive).

Step 1: A less-than ogive uses upper class boundaries against cumulative frequency. Begin at the first lower boundary with zero.

Step 2: Plot the eight labelled points on consistent scales and join them in increasing order, as shown.

Final Answer: Plot

(64.5,0),(69.5,10),(74.5,22),(79.5,43),(84.5,49),(89.5,58),(94.5,62),(99.5,66).

11(a)(iii). Calculate the grouped mode to 3 decimal places.

Step 1: Frequencies are 10,12,21,6,9,4,4. Modal class is 75–79 with lower boundary 74.5 and width 5.

Step 2: $d_1 = 21 - 12 = 9$; $d_2 = 21 - 6 = 15$. Mode = $74.5 + [9/(9 + 15)] \times 5 = 76.375$.

Final Answer: 76.375 marks.

11(b). PT is tangent to a circle of radius 5 cm. P,C,O are collinear, C is on the near circumference and PC = 8 cm. Find PT.

Step 1: $OP = PC + CO = 8 + 5 = 13$ cm. The radius OT is perpendicular to tangent PT.

Step 2: In right triangle OTP, $PT^2 = OP^2 - OT^2 = 13^2 - 5^2 = 144$, so $PT = 12$ cm.

Final Answer: 12 cm.

12(a)(i). A sphere has surface area 113.04 cm^2 . Find its diameter using $\pi = 3.14$.

Step 1: Surface area $= 4\pi r^2$, so $r^2 = 113.04/(4 \times 3.14) = 9$.

Step 2: $r = 3 \text{ cm}$; diameter $= 2r = 6 \text{ cm}$.

Final Answer: 6 cm.

12(a)(ii). 75.360 litres of water fill part of a cylinder of diameter 40 cm. Find the water height.

Step 1: Volume $= 75.360 \times 1000 = 75,360 \text{ cm}^3$; radius $= 20 \text{ cm}$.

Step 2: $V = \pi r^2 h$ gives $h = 75,360/(3.14 \times 20^2) = 75,360/1256 = 60 \text{ cm}$.

Final Answer: 60 cm (using $\pi = 3.14$).

12(b)(i). Find nautical miles between A($18^\circ\text{N}, 12^\circ\text{E}$) and B($65^\circ\text{N}, 12^\circ\text{E}$).

Step 1: Same meridian and same hemisphere give angle $65 - 18 = 47^\circ$.

Step 2: One degree of meridian arc is 60 nautical miles: $47 \times 60 = 2820$.

Final Answer: 2820 nautical miles.

12(b)(ii). Find nautical miles between C($31^\circ\text{S}, 76^\circ\text{W}$) and D($22^\circ\text{N}, 76^\circ\text{W}$).

Step 1: The points cross the equator, so angular separation $= 31 + 22 = 53^\circ$.

Step 2: Distance $= 53 \times 60 = 3180$ nautical miles.

Final Answer: 3180 nautical miles.

13(a)(i). Roza's $R = [[2, 3], [5, -6]]$, Juma's $J = [[-6, 2], [4, 3]]$. Find $2R + 3J$.

Step 1: $2R = [[4, 6], [10, -12]]$ and $3J = [[-18, 6], [12, 9]]$.

Step 2: Add corresponding entries: $[[4 - 18, 6 + 6], [10 + 12, -12 + 9]]$.

Final Answer: $[[-14, 12], [22, -3]]$.

13(a)(ii). Show that the difference between the determinants is ± 1 .

Step 1: $\det R = 2(-6) - 3(5) = -27$. $\det J = (-6)(3) - 2(4) = -26$.

Step 2: Difference $= -27 - (-26) = -1$; the absolute difference is 1.

Final Answer: $\det R - \det J = -1$; reversing the subtraction gives $+1$.

13(b). A 72 m by 88 m farm is enlarged with scale factor 4. Find its new area.

Step 1: Original area = $72 \times 88 = 6336 \text{ m}^2$. A length enlargement of 4 multiplies area by $4^2 = 16$.

Step 2: New area = $6336 \times 16 = 101,376 \text{ m}^2$. Equivalently, new dimensions are 288 m and 352 m.

Final Answer: 101,376 m².

14(a)(i). Quiz score is given by $y = 3x^2$ for x correct answers. Find the score for 8 correct answers.

Step 1: Substitute $x = 8$ into the stated function.

Step 2: $y = 3(8^2) = 3 \times 64 = 192$.

Final Answer: 192 points.

14(a)(ii). Give the domain and range of the inverse of the quiz function $y = 3x^2$.

Step 1: The original input x counts correct answers, so $x \geq 0$; choose the nonnegative inverse $f^{-1}(y) = \sqrt{y/3}$.

Step 2: An inverse's domain is the original range, and its range is the original domain. For integer counts, original outputs are 0,3,12,27,48,...

Final Answer: For quiz counts: inverse domain = $\{3n^2 : n = 0,1,2,\dots\}$, inverse range = $\{0,1,2,\dots\}$. In the continuous nonnegative model, both are $[0,\infty)$.

Study Tip: Over all real x the function $3x^2$ is not one-to-one. The quiz context supplies the nonnegative restriction needed for an inverse function.

14(b)(i). Foods I and II provide vitamins A,B,C per kg as (1,2,3) and (2,2,1), cost Tsh 6000 and 10000. Requirements are at least 10,12,8 units. Formulate a minimum-cost model.

Step 1: Let x and y be kilograms of foods I and II. Multiply each vitamin content by its food quantity and add.

Step 2: At least requires $\geq$ for every nutrient. Food quantities are nonnegative; cost is the weighted sum of kilograms.

Final Answer: Minimise $C = 6000x + 10000y$, subject to $x + 2y \geq 10$; $2x + 2y \geq 12$; $3x + y \geq 8$; $x, y \geq 0$.

14(b)(ii). Find the minimum mixture cost.

Step 1: Feasible lower-bound vertices are (0,8),(1,5),(2,4),(10,0). The intersection of $x + 2y = 10$ and $x + y = 6$ gives (2,4).

Step 2: Costs at the vertices are 80,000; 56,000; 52,000; 60,000 respectively. The smallest is 52,000.

Step 3: Check vitamins at (2,4): $A = 2 + 8 = 10$; $B = 4 + 8 = 12$; $C = 6 + 4 = 10 \geq 8$. All requirements are satisfied.

Final Answer: Tsh 52,000, using 2 kg of food I and 4 kg of food II.

Section Summary: Recover frequencies from cumulative totals before averaging or estimating a mode. Area scales use squared length factors, and inverse functions need an appropriate input domain.